

Graphical Solution

Graphical Solution of LP Models

- Graphical solution is limited to linear programming models containing *only two decision variables* (can be used with three variables but only with great difficulty).
- Used to identify the optimal solution to the model.
- Plot one decision variable on the **horizontal (X) axis**, and another variable on the **vertical (Y) axis**
- The nonnegativity constraints: first (or positive) quadrant of a graph.
- Graphical methods provide *visualization of how* a solution for a linear programming problem is obtained.

Coordinate Axes

Graphical Solution of Maximization Model

X2 is mugs

Maximize $Z = \$40x_1 + \$50x_2$

subject to: $1x_1 + 2x_2 \leq 40$

$4x_1 + 3x_2 \leq 120$

$x_1, x_2 \geq 0$

Let x_1 be represented on X axis

Let x_2 be represented on Y axis

Constraint 1

1. Convert it to equality

$$x_1 + 2x_2 = 40$$

2. Coordinates

Put $x_1 = 0$; $0 + 2x_2 = 40$; $x_2 = 20$; $(0, 20)$

Put $x_2 = 0$; $x_1 + 2 \cdot 0 = 40$; $x_1 = 40$; $(40, 0)$

Constraint 2

$$4x_1 + 3x_2 = 120$$

$x_1 = 0$; $x_2 = 40$

$x_2 = 0$; $x_1 = 30$

$(0, 40)$, $(30, 0)$

X1 is bowls

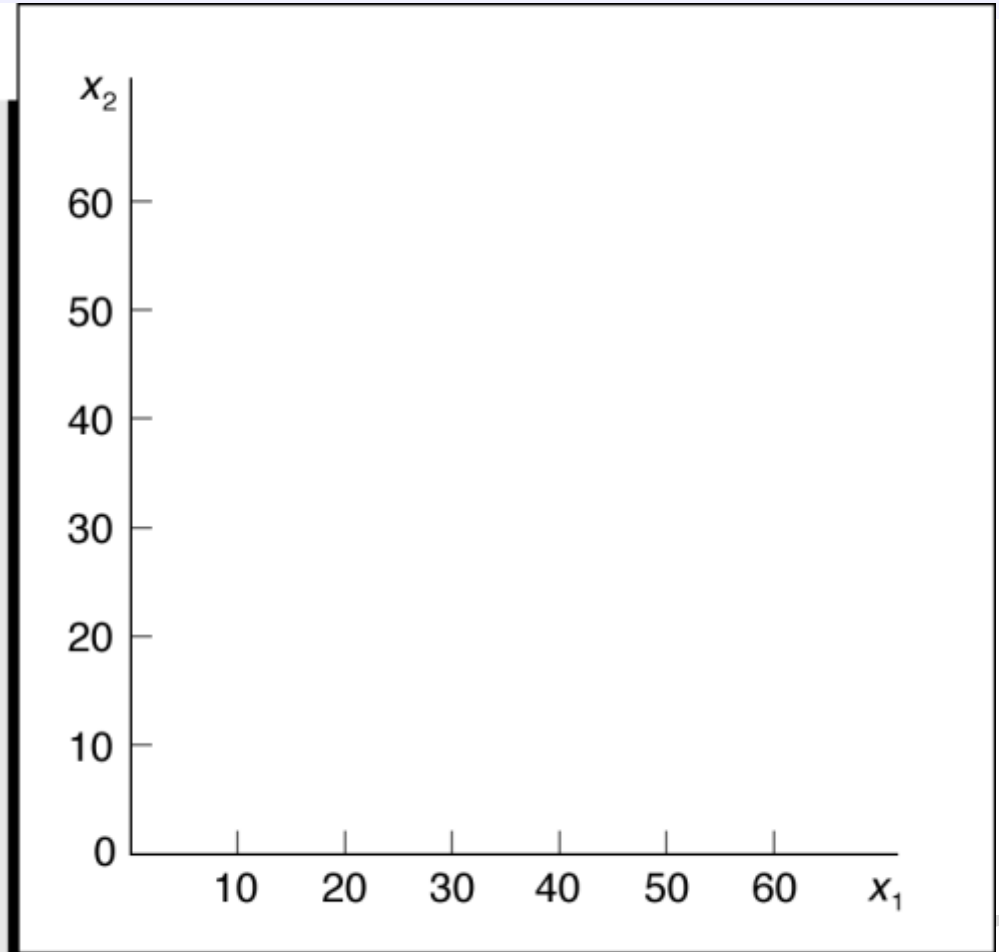


Figure 2.2 Coordinates for Graphical Analysis

Labor Constraint

Graphical Solution of Maximization Model

Maximize $Z = \$40x_1 + \$50x_2$
subject to:
 $1x_1 + 2x_2 \leq 40$
 $4x_1 + 3x_2 \leq 120$
 $x_1, x_2 \geq 0$

Constraint 1

1. Convert it to equality

$$x_1 + 2x_2 = 40$$

2. Coordinates

Put $x_1=0$; $0+2x_2=40$; $x_2=20$; $(0, 20)$

Put $x_2=0$; $x_1+2*0=40$; $x_1=40$; $(40,0)$

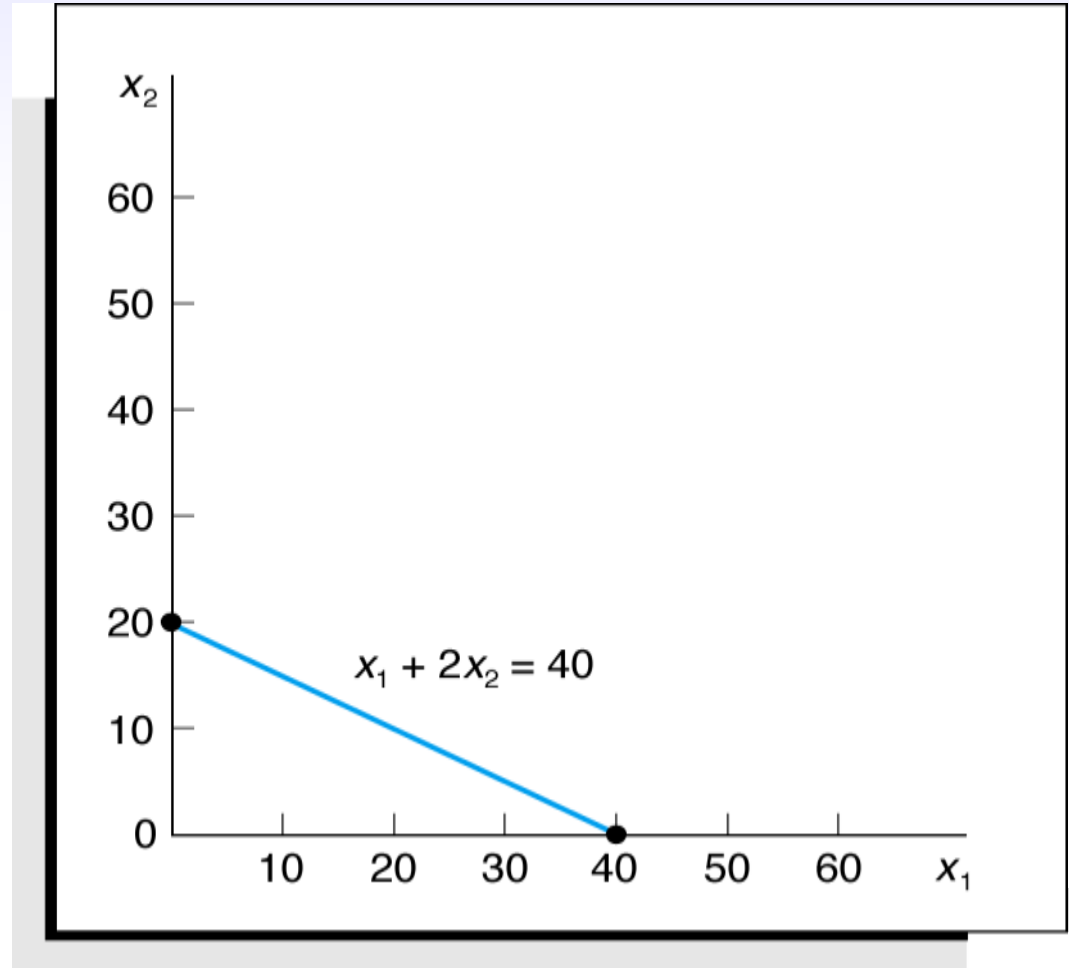


Figure 2.3 Graph of Labor Constraint

Labor Constraint Area

Graphical Solution of Maximization Model

Maximize $Z = \$40x_1 + \$50x_2$
subject to:
 $1x_1 + 2x_2 \leq 40$
 $4x_1 + 3x_2 \leq 120$
 $x_1, x_2 \geq 0$

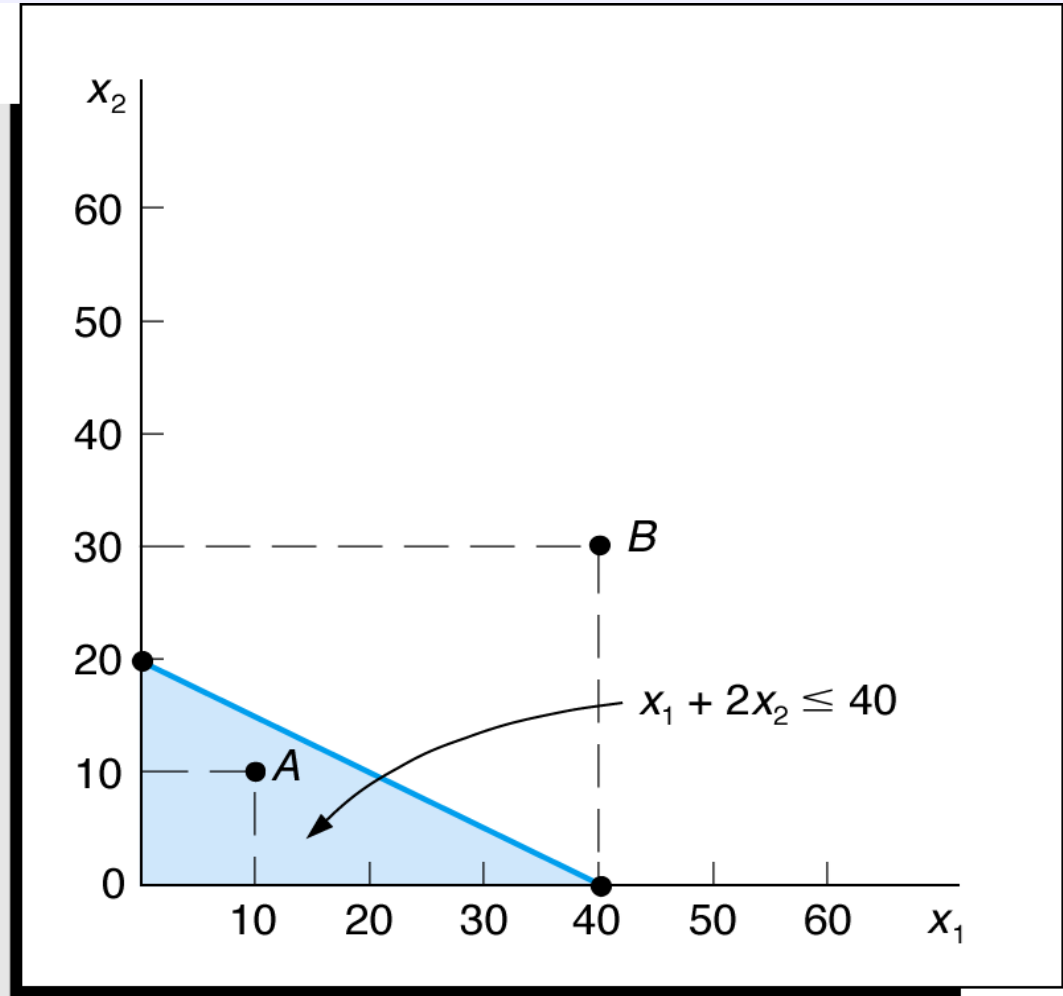


Figure 2.4 Labor Constraint Area

Clay Constraint Area

Graphical Solution of Maximization Model

Maximize $Z = \$40x_1 + \$50x_2$
subject to:
 $1x_1 + 2x_2 \leq 40$
 $4x_1 + 3x_2 \leq 120$
 $x_1, x_2 \geq 0$

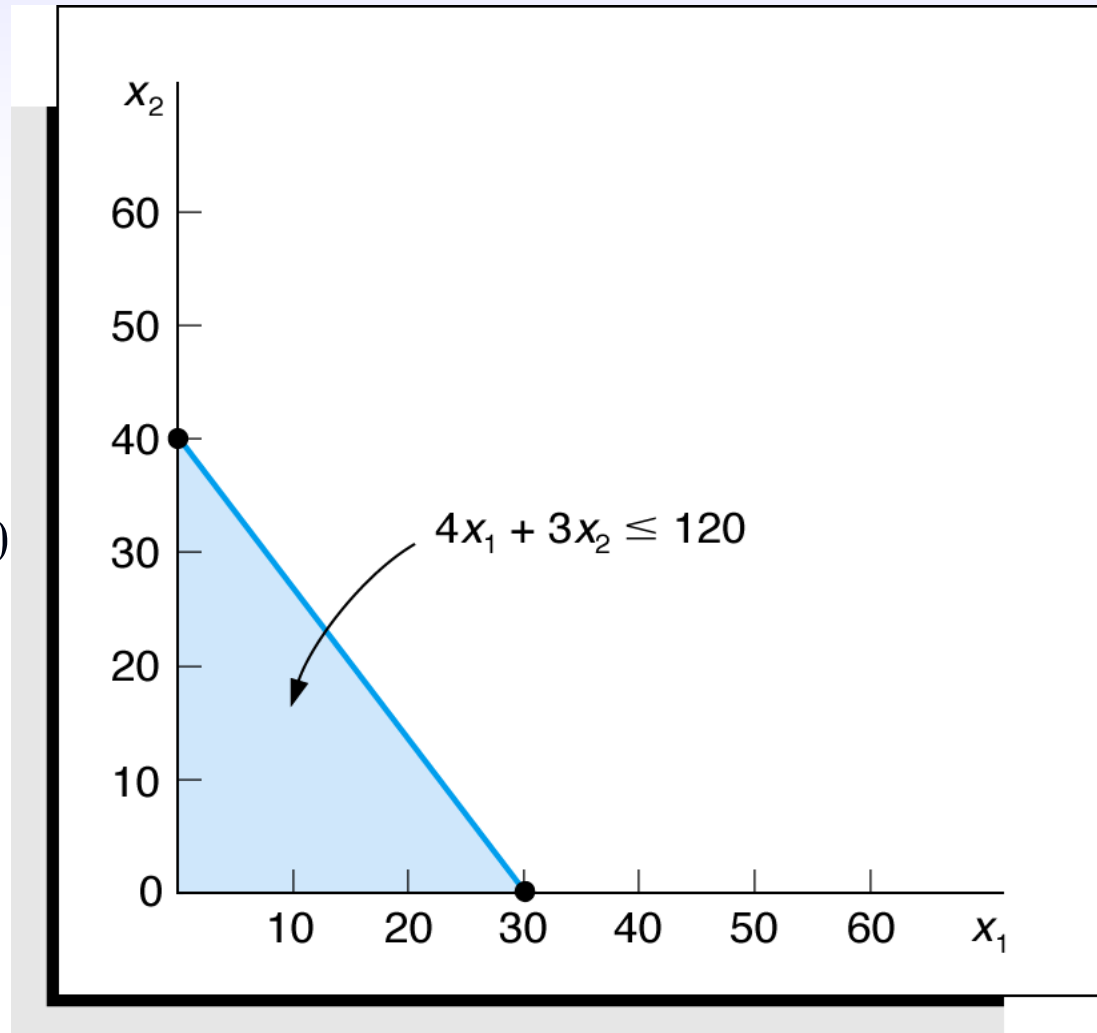


Figure 2.5 Clay Constraint Area

Both Constraints

Graphical Solution of Maximization Model

Feasible Region

- ✓ The feasible region of an LP problem consists of those points that simultaneously satisfy all constraints
- ✓ Region where all the problem's constraints overlap.

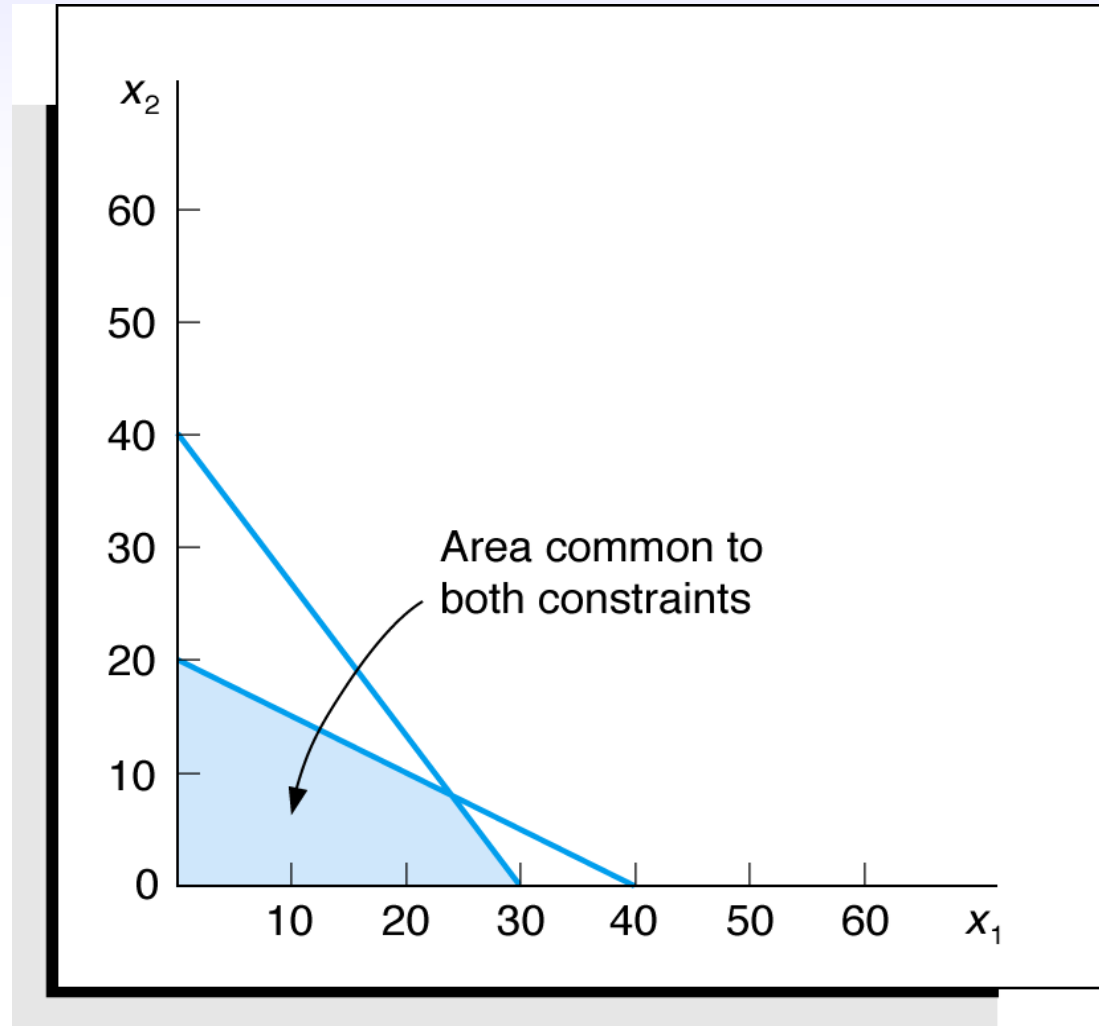


Figure 2.6 Graph of Both Model Constraints

Both Constraints

Graphical Solution of Maximization Model

Corner Points

Where two or more constraints intersect, are called corner points, or extreme points

Corner point property

This property states that an optimal solution to an LP problem will always occur at a corner point of the feasible region.

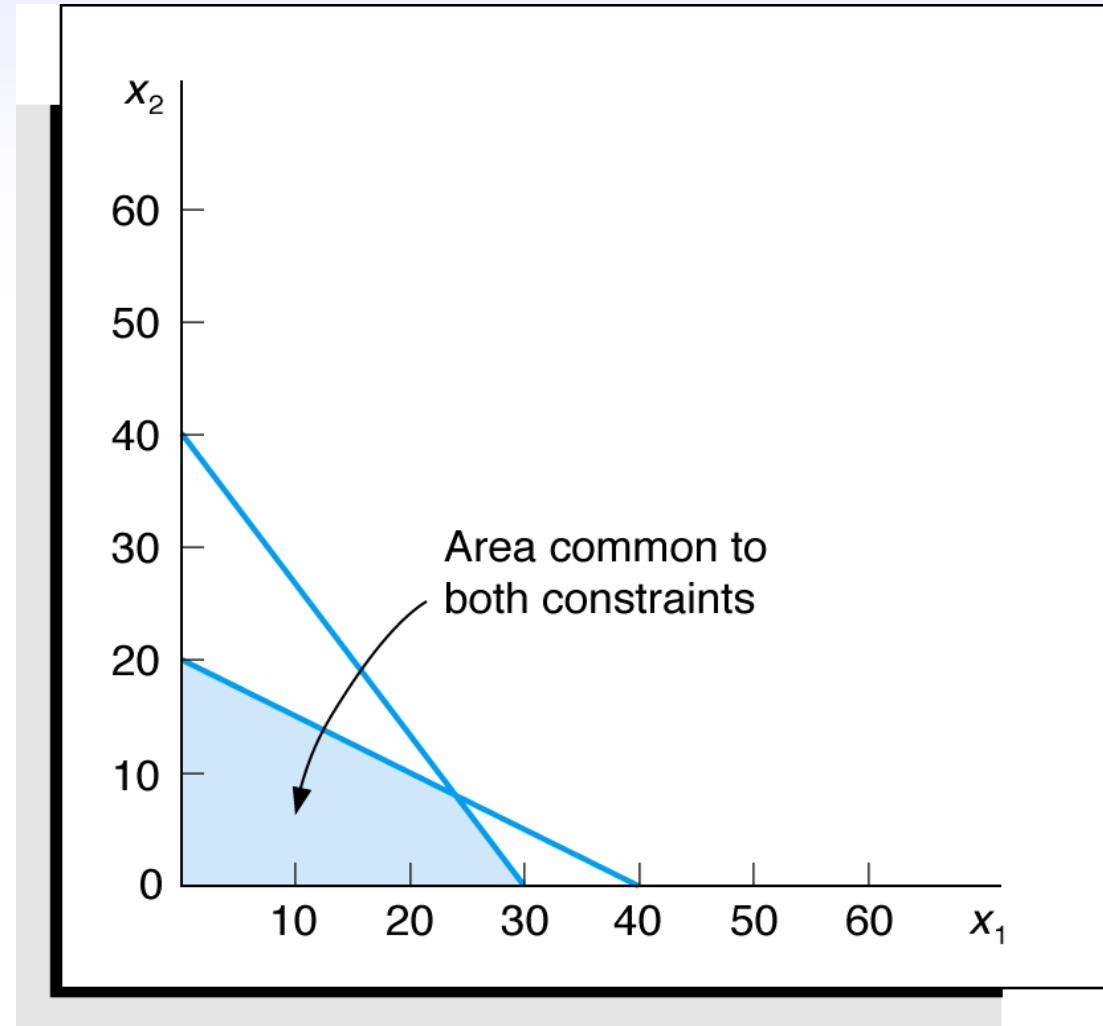


Figure 2.6 Graph of Both Model Constraints

Both Constraints

Graphical Solution of Maximization Model

Maximize $Z = \$40x_1 + \$50x_2$

subject to: $1x_1 + 2x_2 \leq 40$

$4x_1 + 3x_2 \leq 120$

$x_1, x_2 \geq 0$

$x_1 + 2x_2 = 40$ i * 4

$-(4x_1 + 3x_2 = 120)$ ii

$5x_2 = 40$; $x_2 = 8$

Put x_2 in eq. 1

$x_1 + 2*8 = 40$

$x_1 = 40 - 16 = 24$

Corner points: A(0,20) B(24,8)

C(30,0)

$Z(A) = 40*0 + 20 * 50 = \1000

$Z(B) = 40*24 + 50*8 = \$1360$

$Z(C) = 40*30 + 50*0 = \$1200$

For profit maximization we need to produce 24 units of x_1 and 8 units of x_2

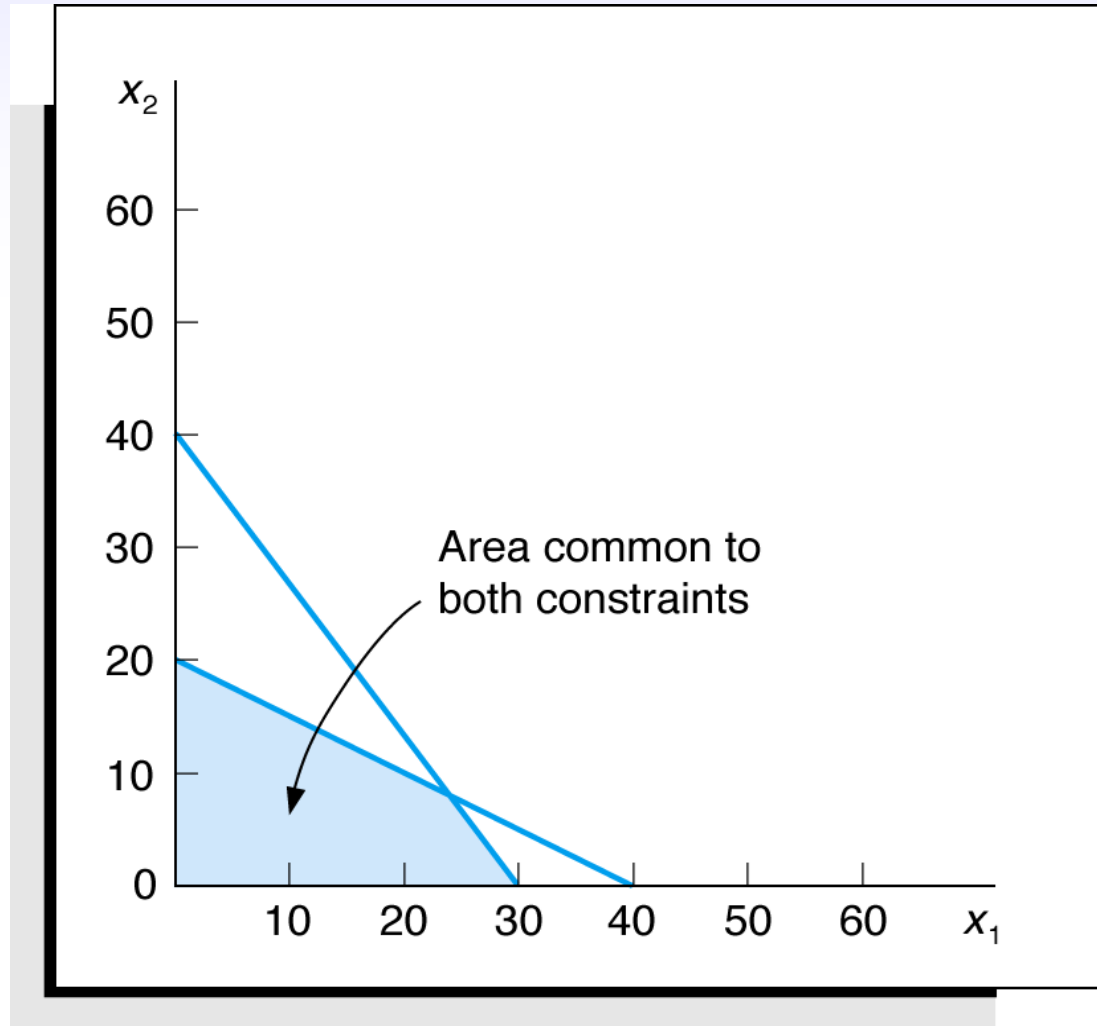


Figure 2.6 Graph of Both Model Constraints

Optimal Solution Coordinates

Graphical Solution of Maximization Model (6 of 12)

Maximize $Z = \$40x_1 + \$50x_2$
subject to:
 $1x_1 + 2x_2 \leq 40$
 $4x_1 + 3x_2 \leq 120$
 $x_1, x_2 \geq 0$

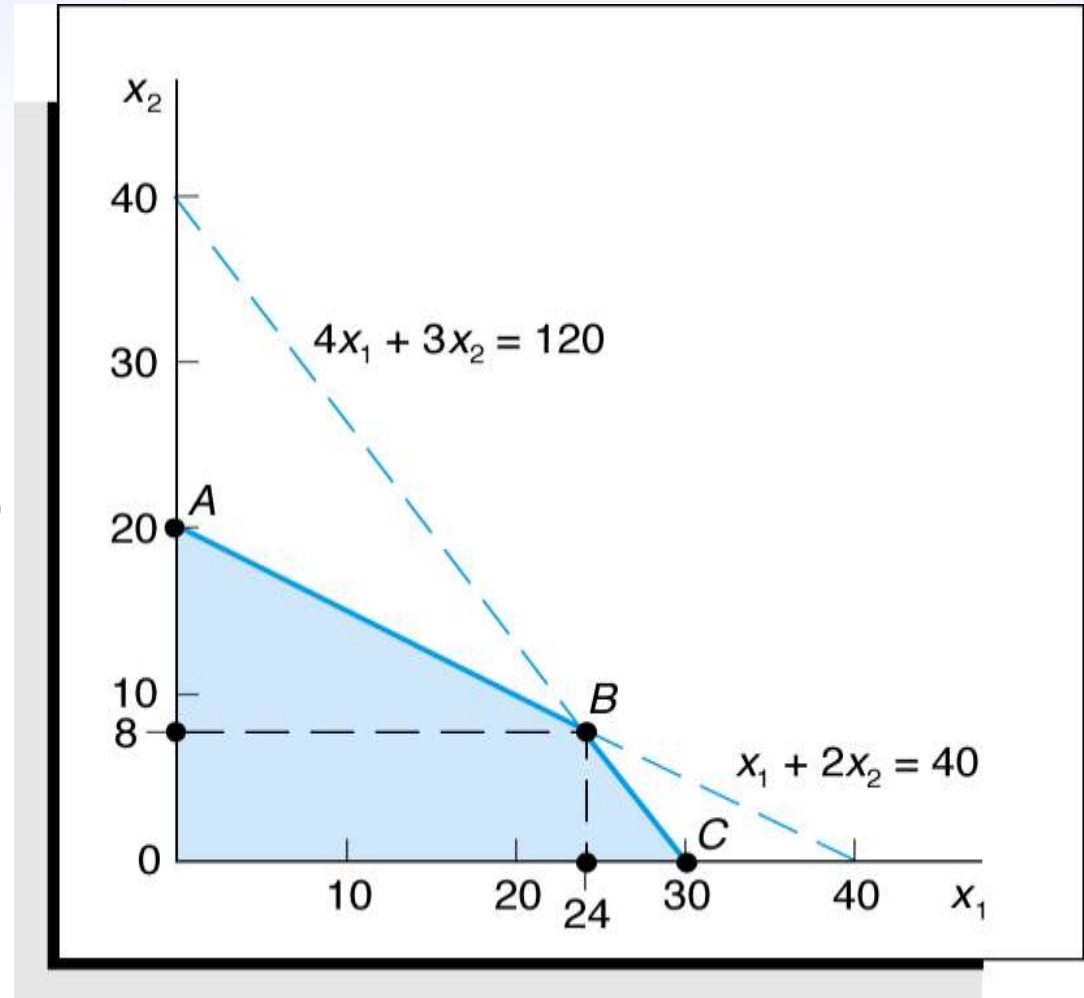


Figure 2.11 Optimal Solution Coordinates

Extreme (Corner) Point Solutions

Graphical Solution of Maximization Model (7 of 12)

Maximize $Z = \$40x_1 + \$50x_2$
subject to:
 $1x_1 + 2x_2 \leq 40$
 $4x_1 + 3x_2 \leq 120$
 $x_1, x_2 \geq 0$

Optimal Solution at point B

That is $x_1 = 24$ & $x_2 = 8$

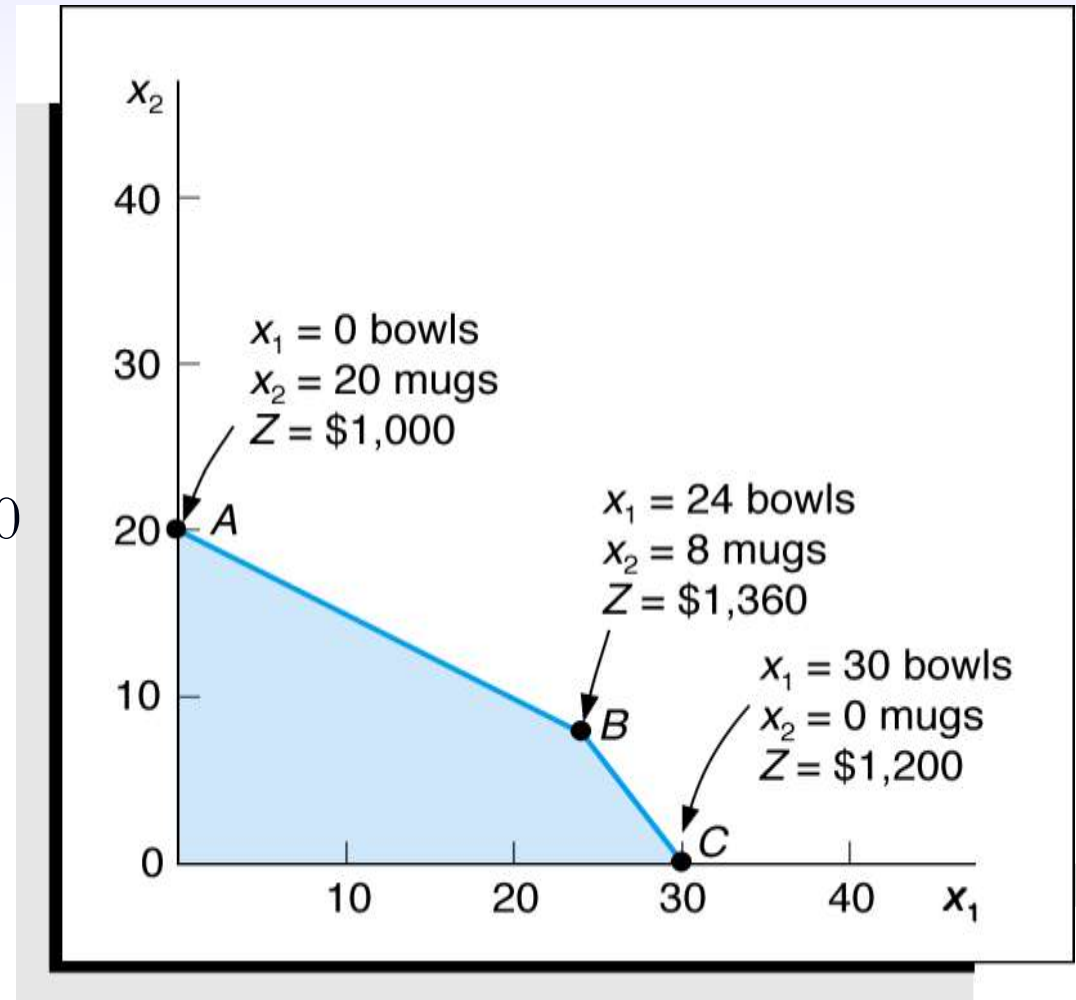


Figure 2.12 Solutions at All Corner Points

LP Model Formulation – Minimization (1 of 8)

- Two brands of fertilizer available - Super-gro, Crop-quick.
- Field requires at least 16 pounds of nitrogen and 24 pounds of phosphate.
- Super-gro costs \$6 per bag, Crop-quick \$3 per bag.
- Problem: How much of each brand to purchase to minimize total cost of fertilizer given following data ?

Brand	Chemical Contribution	
	Nitrogen (lb/bag)	Phosphate (lb/bag)
Super-gro	2	4
Crop-quick	4	3

LP Model Formulation – Minimization (2 of 8)

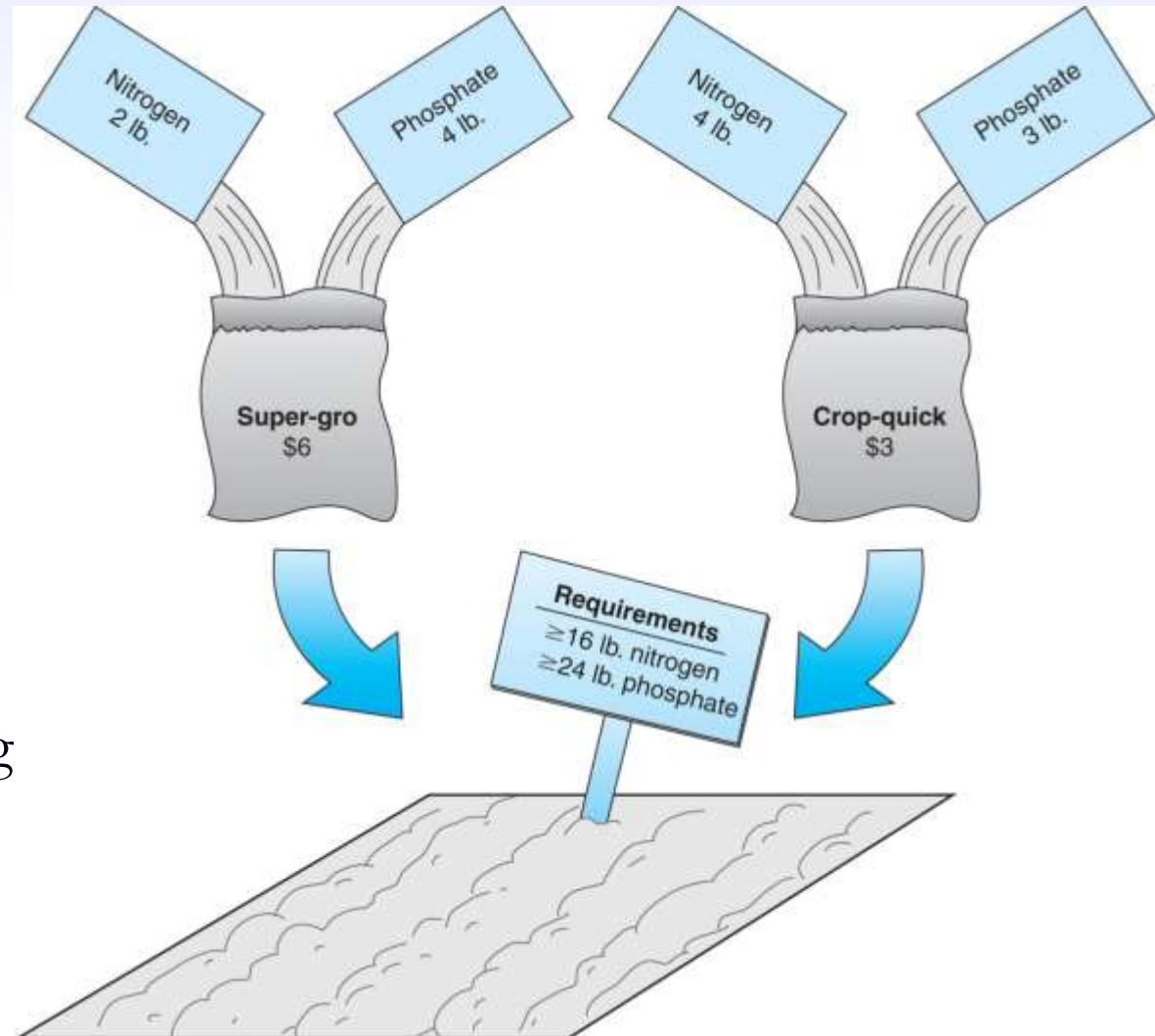


Figure 2.15 Fertilizing farmer's field

LP Model Formulation – Minimization (3 of 8)

Decision Variables:

x_1 = bags of Super-gro

x_2 = bags of Crop-quick

The Objective Function:

Minimize $Z = \$6x_1 + \$3x_2$

Where: $\$6x_1$ = cost of bags of Super-Gro

$\$3x_2$ = cost of bags of Crop-Quick

Model Constraints:

$2x_1 + 4x_2 \geq 16$ lb (nitrogen constraint)

$4x_1 + 3x_2 \geq 24$ lb (phosphate constraint)

$x_1, x_2 \geq 0$ (non-negativity constraint)

Constraint Graph – Minimization (4 of 8)

Minimize $Z = \$6x_1 + \$3x_2$
subject to:
 $2x_1 + 4x_2 \geq 16$
 $4x_1 + 3x_2 \geq 24$
 $x_1, x_2 \geq 0$

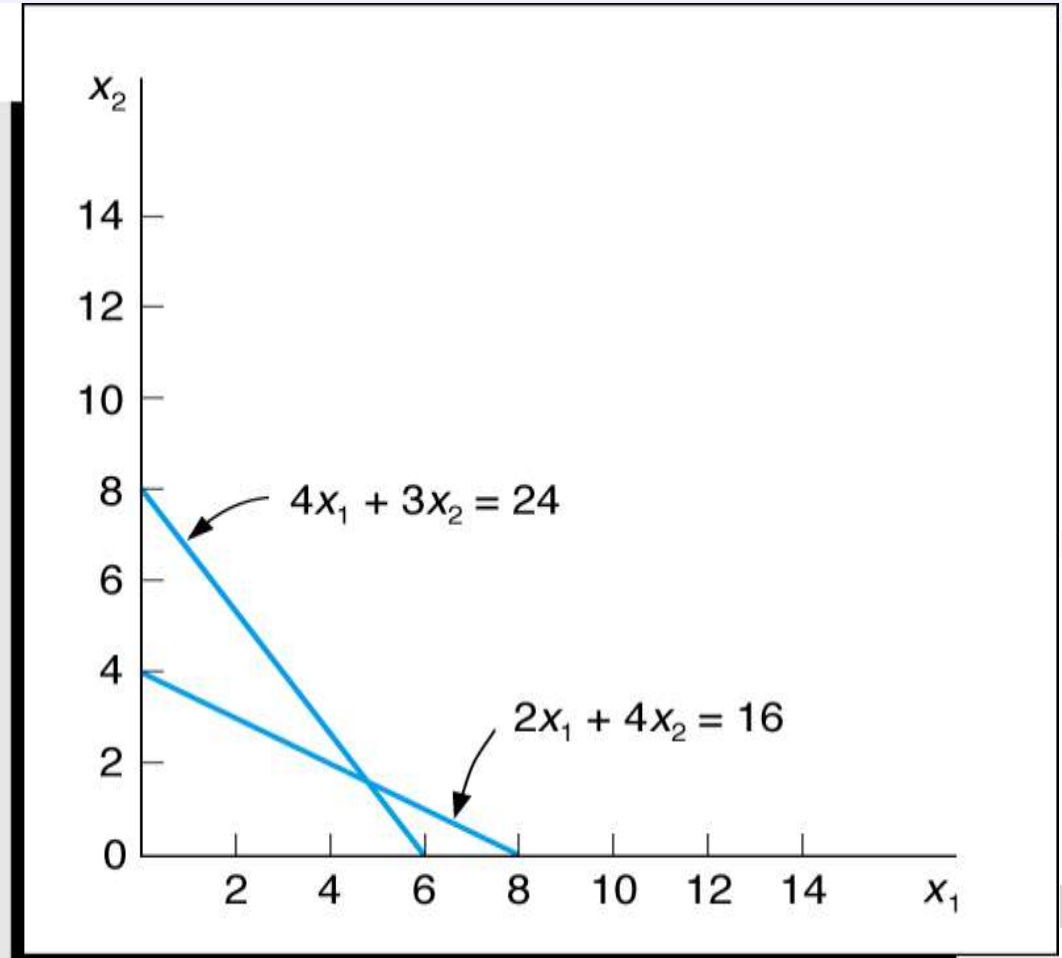


Figure 2.16 Graph of Both Model Constraints

Feasible Region– Minimization (5 of 8)

Minimize $Z = \$6x_1 + \$3x_2$
subject to: $2x_1 + 4x_2 \geq 16$
 $4x_1 + 3x_2 \geq 24$
 $x_1, x_2 \geq 0$

A(0,8), B(4.8,1.6), C(8,0)

$$Z(A) = 6*0 + 8*3 = 24$$

$$Z(B) = 6*4.8 + 3*1.6 = 33.6$$

$$Z(C) = 6*8 + 3*0 = 48$$

$x_1 = 0$; $x_2 = 8$ units

$$\begin{array}{r} (2x_1 + 4x_2 = 16) * 2 \\ -(4x_1 + 3x_2 = 24) \\ \hline 0 + 5x_2 = 8 \end{array}$$

$$x_2 = 1.6$$

$$2x_1 + 4*1.6 = 16; x_1 = 4.8$$

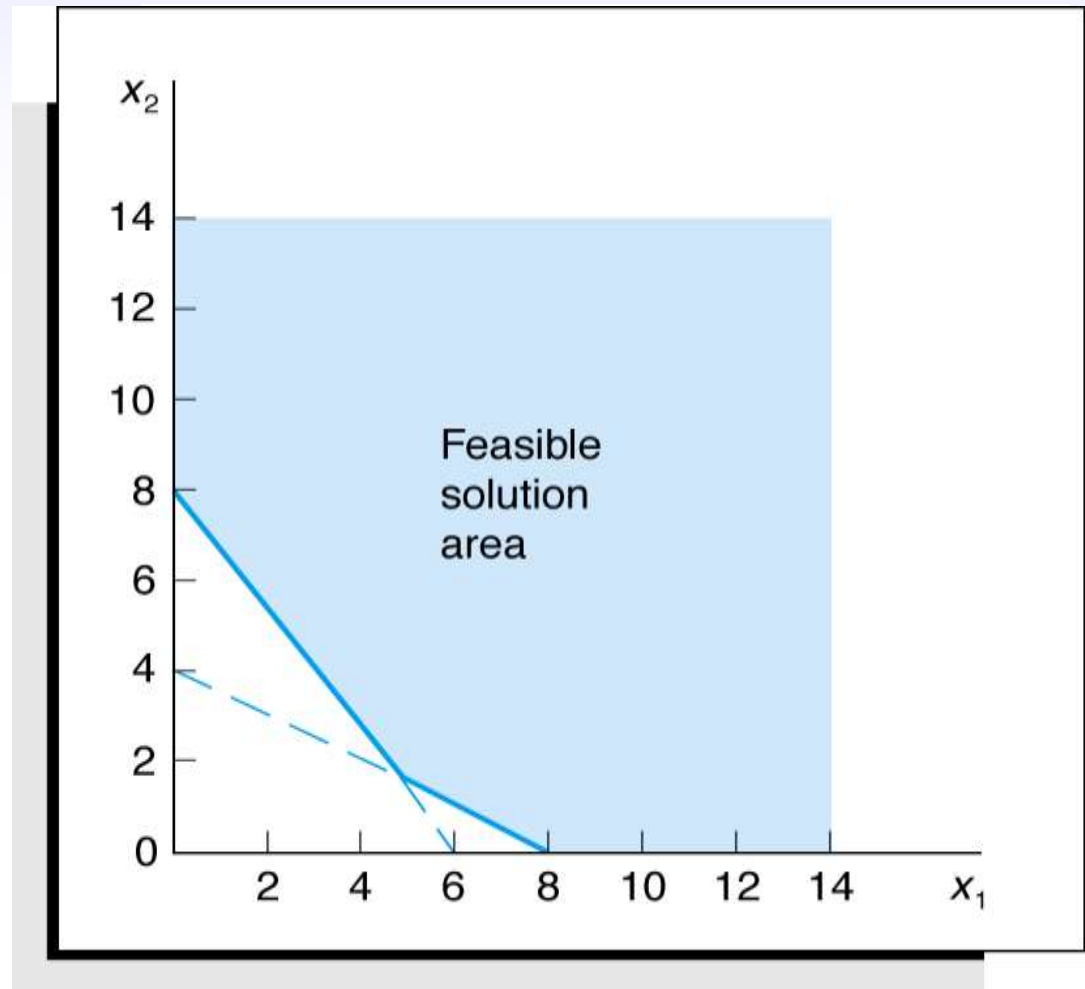


Figure 2.17 Feasible Solution Area

Optimal Solution Point – Minimization (6 of 8)

Minimize $Z = \$6x_1 + \$3x_2$
subject to: $2x_1 + 4x_2 \geq 16$
 $4x_1 + 3x_2 \geq 24$
 $x_1, x_2 \geq 0$

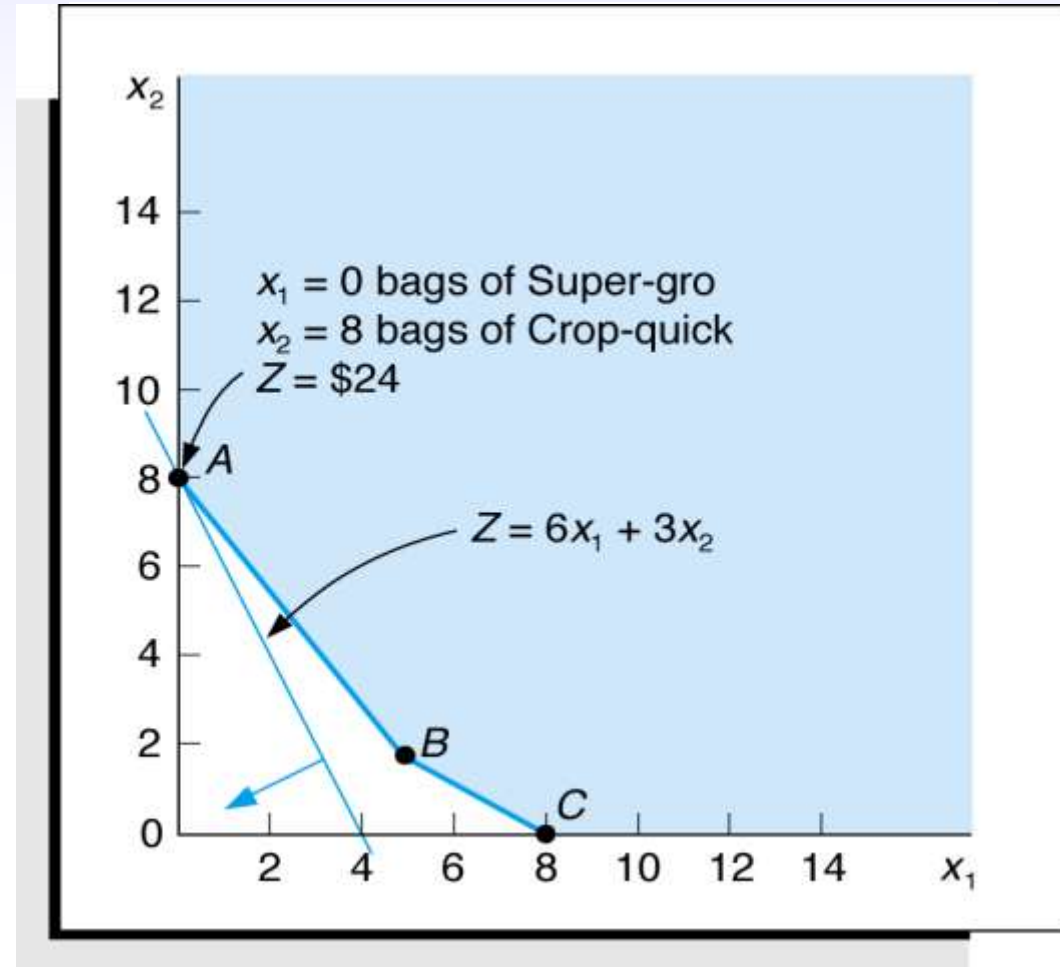


Figure 2.18 Optimum Solution Point

Irregular Types of Linear Programming Problems

For some linear programming models, the general rules do not apply.

■ Special Situations in Solving Linear Programming Problems

Special types of problems include those with:

- Redundant Constraint
- Infeasible solutions
- Unbounded solutions
- Multiple optimal solutions

An Infeasible Problem

Infeasibility is a condition that arises when no single solution satisfies all of an LP problem's constraints.

That is, no feasible solution region exists.

Every possible solution **violates** at least one constraint:

$$\begin{aligned} \text{Maximize } Z &= 5x_1 + 3x_2 \\ \text{subject to: } &4x_1 + 2x_2 \leq 8 \\ &x_1 \geq 4 \\ &x_2 \geq 6 \\ &x_1, x_2 \geq 0 \end{aligned}$$

Constraint 1;

$$x_1=0, x_2=4; (0,4)$$

$$x_2=0, x_1=2; (2,0)$$

Constraint 2;

$$x_1=4$$

Constraint 3;

$$x_2=6$$

An Infeasible Problem

Every possible solution
violates at least one constraint:

Maximize $Z = 5x_1 + 3x_2$
subject to: $4x_1 + 2x_2 \leq 8$
 $x_1 \geq 4$
 $x_2 \geq 6$
 $x_1, x_2 \geq 0$

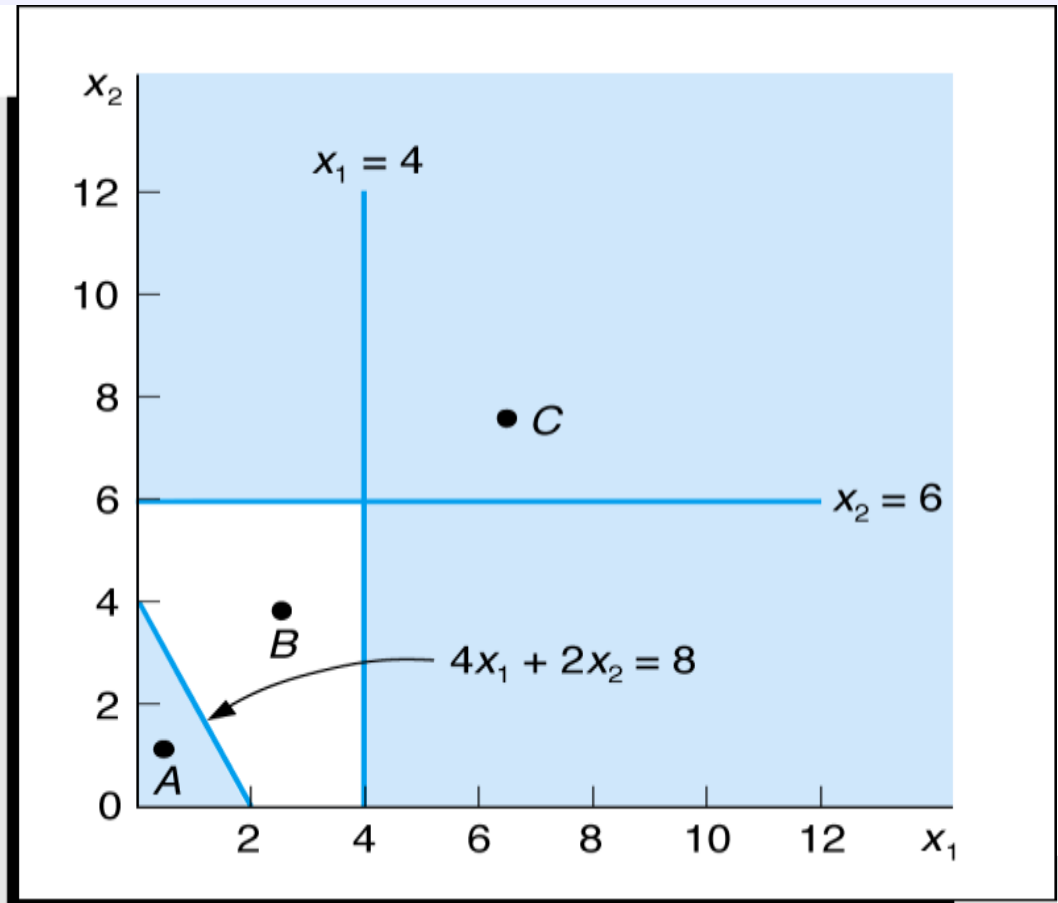


Figure 2.21 Graph of an Infeasible Problem

An Unbounded Problem

If the feasible region is unbounded in one or more directions, depending on the objective function, the model may or may not have a finite solution.

Value of the objective function increases indefinitely:

Maximize $Z = 4x_1 + 2x_2$
subject to: $x_1 \geq 4$
 $x_2 \leq 2$
 $x_1, x_2 \geq 0$

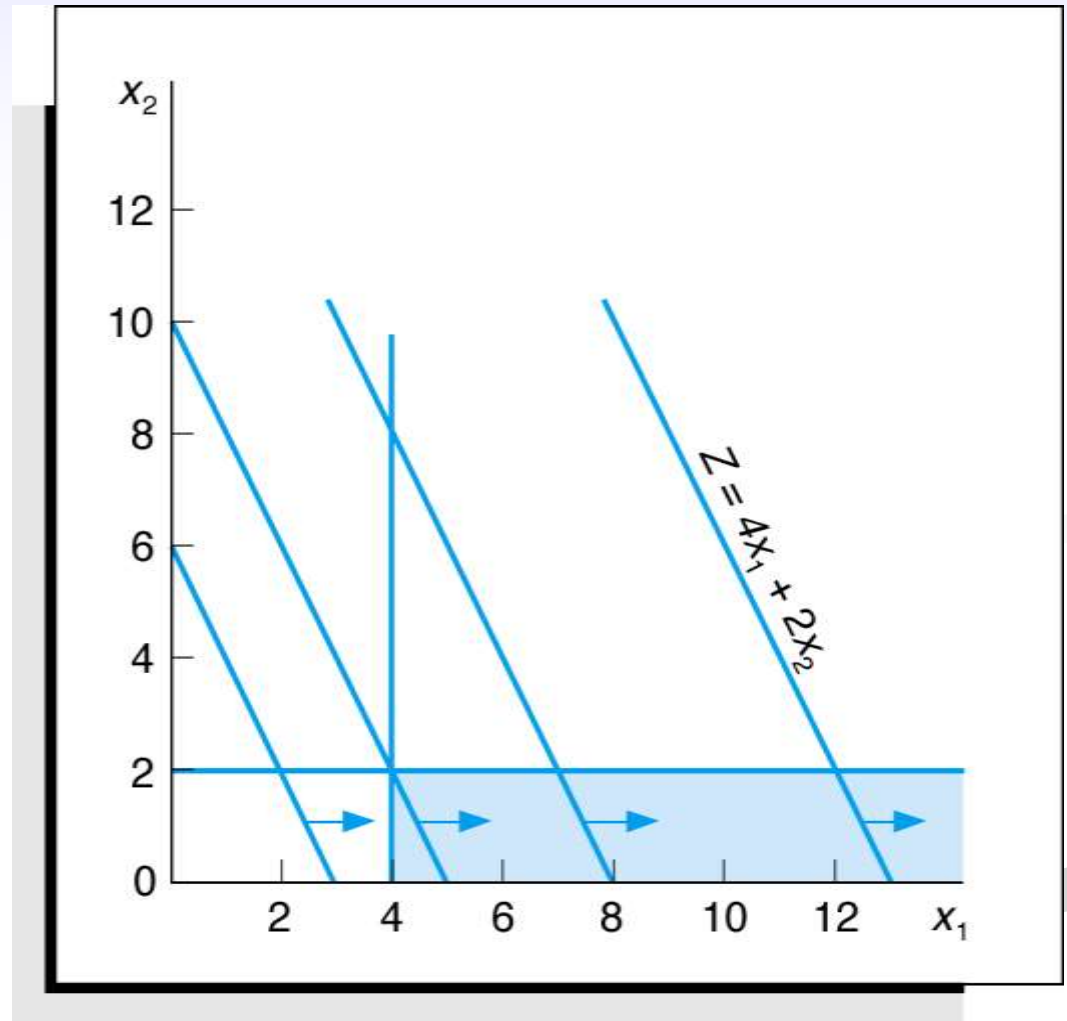


Figure 2.22 Graph of an Unbounded Problem

A Multiple Solution Problem

Multiple optimal solutions

E.g.,

$$\text{Max } Z = 3X_1 + 2X_2$$

$$X_1 \leq 4$$

$$2X_2 \leq 12$$

$$3X_1 + 2X_2 \leq 18$$

$$X_1, X_2 \geq 0$$

Characteristics of Linear Programming Problems

- A decision amongst alternative courses of action is required.
- The decision is represented in the model by **decision variables**.
- The problem encompasses a goal, expressed as an **objective function**, that the decision maker wants to achieve.
- Restrictions (represented by **constraints**) exist that limit the extent of achievement of the objective.
- The objective and constraints must be definable by **linear** mathematical functional relationships.

Problem Statement

Example Problem No. 1 (1 of 3)

- Hot dog mixture in 1000-pound batches.
- Two ingredients, chicken (\$3/lb) and mutton (\$5/lb).
- Recipe requirements:
 - at least 500 pounds of “chicken”
 - at least 200 pounds of “mutton”
- Ratio of chicken to mutton must be at least 2 to 1.
- Determine optimal mixture of ingredients that will minimize costs.

Solution

Example Problem No. 1 (2 of 3)

Step 1:

Identify decision variables.

x_1 = lb of chicken in mixture

x_2 = lb of mutton in mixture

Step 2:

Formulate the objective function.

Minimize $Z = \$3x_1 + \$5x_2$

where Z = cost per 1,000-lb batch

$\$3x_1$ = cost of chicken

$\$5x_2$ = cost of mutton

Solution

Example Problem No. 1 (3 of 3)

Step 3:

Establish Model Constraints

$$x_1 + x_2 = 1,000 \text{ lb}$$

$$x_1 \geq 500 \text{ lb of chicken}$$

$$x_2 \geq 200 \text{ lb of mutton}$$

$$x_1/x_2 \geq 2/1 \text{ or } x_1 - 2x_2 \geq 0$$

$$x_1, x_2 \geq 0$$

The Model: Minimize $Z = \$3x_1 + 5x_2$
subject to: $x_1 + x_2 = 1,000 \text{ lb}$
 $x_1 \geq 500$
 $x_2 \geq 200$
 $x_1 - 2x_2 \geq 0$
 $x_1, x_2 \geq 0$

Example Problem No. 2 (1 of 3)

reading exercise on page 58

Solve the following model graphically:

Maximize $Z = 4x_1 + 5x_2$
subject to: $x_1 + 2x_2 \leq 10$
 $6x_1 + 6x_2 \leq 36$
 $x_1 \leq 4$
 $x_1, x_2 \geq 0$

Step 1: Plot the constraints as equations

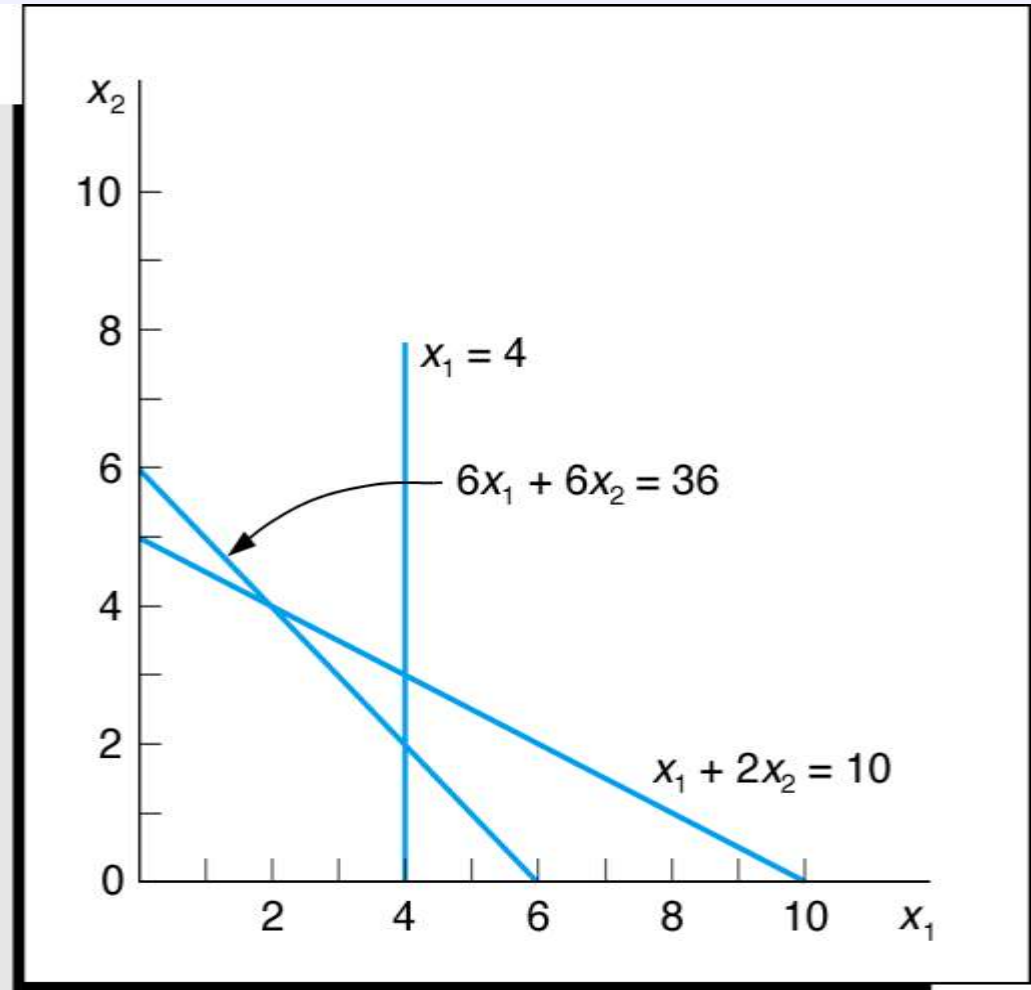


Figure 2.23 Constraint Equations

Example Problem No. 2 (2 of 3)

Maximize $Z = 4x_1 + 5x_2$
subject to: $x_1 + 2x_2 \leq 10$
 $6x_1 + 6x_2 \leq 36$
 $x_1 \leq 4$
 $x_1, x_2 \geq 0$

Step 2: Determine the feasible solution space

For point A(0,5), B (2,4), C(4,2), D(4,0)

$$Z(A) = 4 \cdot 0 + 5 \cdot 5 = 25$$

$$Z(B) = 4 \cdot 2 + 5 \cdot 4 = 28$$

$$Z(C) = 4 \cdot 4 + 5 \cdot 2 = 26$$

$$Z(D) = 4 \cdot 4 + 5 \cdot 0 = 16$$

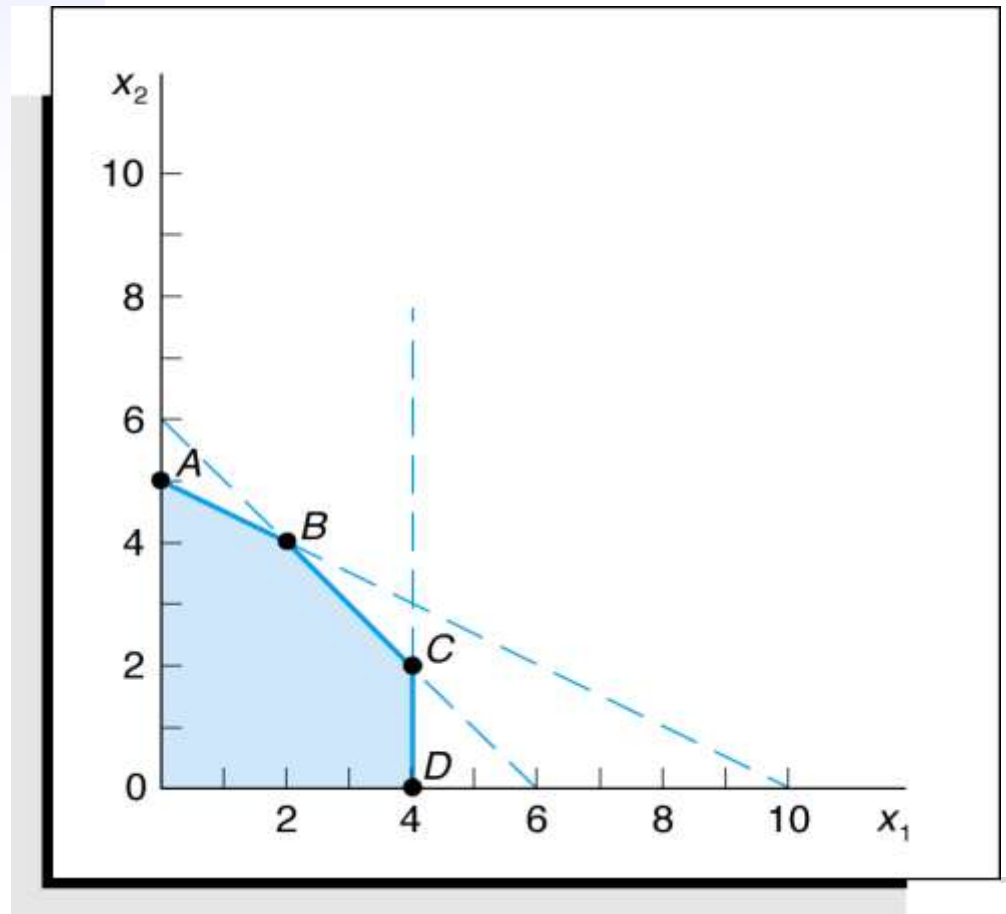


Figure 2.24 Feasible Solution Space and Extreme Points

Example Problem No. 2 (3 of 3)

Maximize $Z = 4x_1 + 5x_2$
subject to: $x_1 + 2x_2 \leq 10$
 $6x_1 + 6x_2 \leq 36$
 $x_1 \leq 4$
 $x_1, x_2 \geq 0$

Step 3 and 4: Determine the solution points and optimal solution

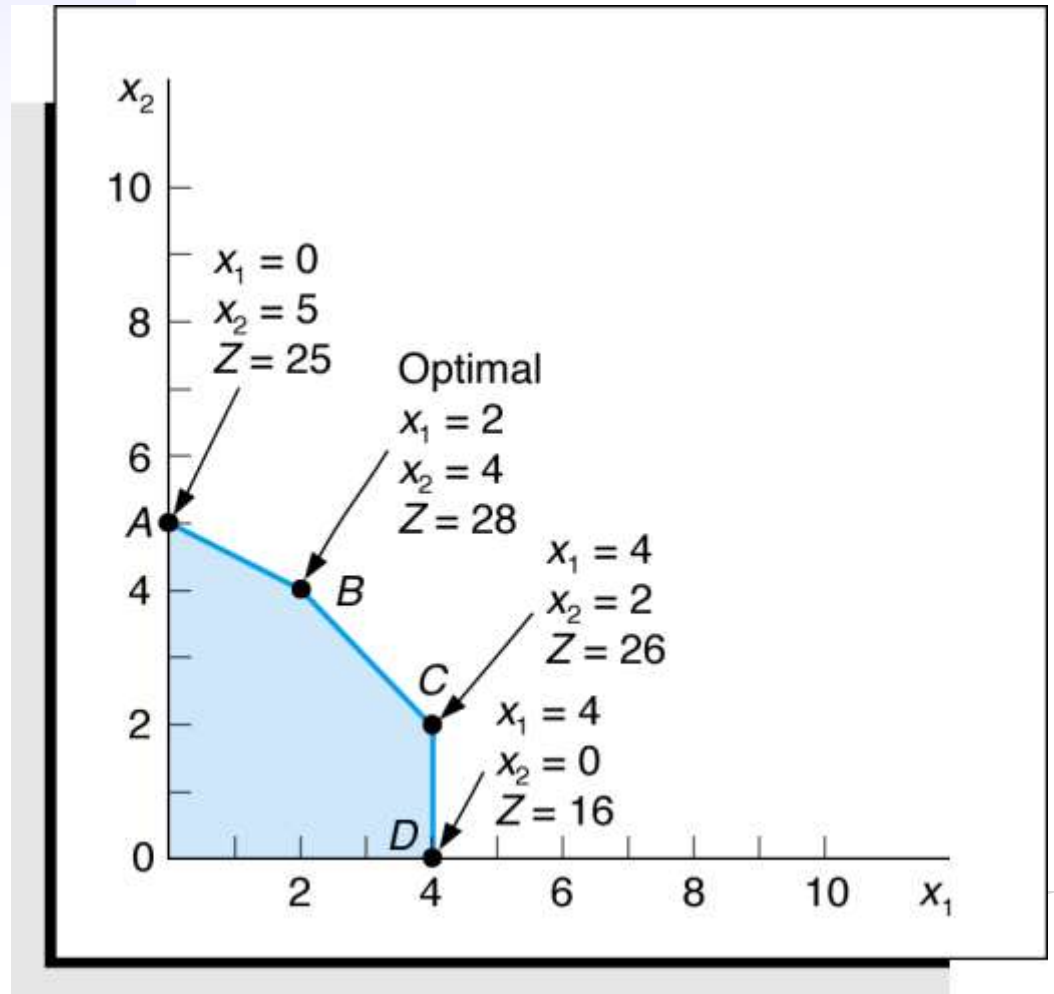


Figure 2.25 Optimal Solution Point

Problem

Reddy Mikks produces both interior and exterior paints from two raw materials, $M1$ and $M2$. The following table provides the basic data of the problem:

	Tons of raw material per ton of		Maximum daily availability (tons)
	<i>Exterior paint</i>	<i>Interior paint</i>	
Raw material, $M1$	6	4	24
Raw material, $M2$	1	2	6
Profit per ton (\$1000)	5	4	

A market survey indicates that the daily demand for interior paint cannot exceed that for exterior paint by more than 1 ton. Also, the maximum daily demand for interior paint is 2 tons.

Reddy Mikks wants to determine the optimum (best) product mix of interior and exterior paints that maximizes the total daily profit.

OR model has three basic components.

Solution

x_1 = Tons produced daily of exterior paint

x_2 = Tons produced daily of interior paint

$$\text{Maximize } z = 5x_1 + 4x_2$$

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

Problem

Example 2.2. An electronic company is engaged in the production of two components C_1 and C_2 used in T.V. sets. Each unit of C_1 costs the company Rs. 25 in wages and Rs. 25 in material, while each unit of C_2 costs the company Rs. 125 in wages and Rs. 75 in material. The company sells both products on one-period credit terms, but the company's labour and material expenses must be paid in cash. The selling price of C_1 is Rs. 150 per unit and of C_2 it is Rs. 350 per unit. Because of the strong monopoly of the company for these components, it is assumed that the company can sell at the prevailing prices as many units as it produces. The company's production capacity is, however, limited by two considerations. First, at the beginning of period 1, the company has an initial balance of Rs. 20,000 (cash plus bank credit plus collections from past credit sales). Second, the company has available in each period 4,000 hours of machine time and 2,800 hours of assembly time. The production of each C_1 requires 6 hours of machine time and 4 hours of assembly time, whereas the production of each C_2 requires 4 hours of machine time and 6 hours of assembly time. Formulate this problem as an LP model so as to maximize the total profit to the company.

Solution

Solution. By designating the number of units of components C_1 and C_2 to be produced as x_1 and x_2 , the data of the given problem can be summarized as follows :

Decision Variables	Components	Budget (Rs.)	Machine Time	Assembly Time	Selling Price (Rs.)	Cost Price (Rs.)
x_1	C_1	50/unit	6 hrs/unit	4 hrs/unit	150	50
x_2	C_2	200/unit	4 hrs/unit	6 hrs/unit	350	200
Total Availability		20,000	4,000 hrs.	2,800 hrs.		

The appropriate mathematical formulation of the above problem as *LP model* is :

Maximize (total profit) $Z = \text{Selling price} - \text{Cost price}$

$$= (150 - 50) x_1 + (350 - 200) x_2 = 100x_1 + 150x_2$$

Subject to the constraints

$$50x_1 + 200 x_2 \leq 20,000 \text{ (Total budget constraint)}$$

$$\left. \begin{array}{l} 6x_1 + 4x_2 \leq 4,000 \\ 4x_1 + 6x_2 \leq 2,800 \end{array} \right\} \text{ (Production time constraint)}$$

$$x_1, x_2 \geq 0$$